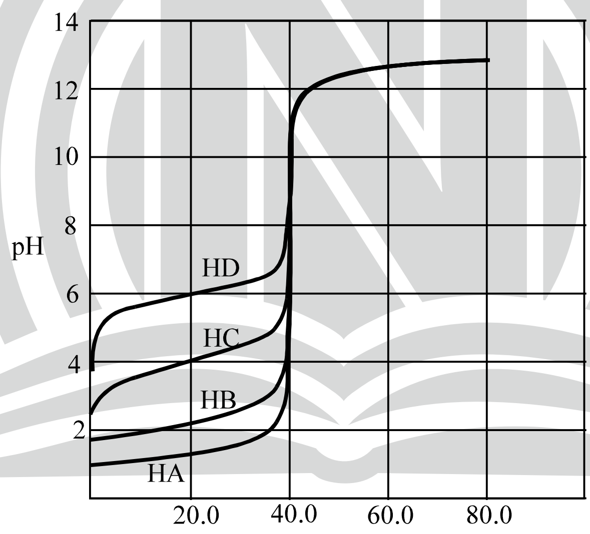
SUBJECT: CHEMISTRY **DATE: 15-03-2024**

COMPREHENSION TYPE:

Paragraph # 1 for (Ques. Nos. 1 to 2)

Following is the titration cure of four acids HA, HB, HC, HD (4 m moles each) titrated against strong base NaOH (0.1 M). Carefully study the graph answer Question No. (13) and (14)



1. What is the equilibrium constant for the reaction

HC (aq) + NaD(aq) $\rightleftharpoons$ HD(aq) + NaC(aq)

(A) $K = 10^{-10}$ (B) $K = 10^{2}$ (C) $K = 10^{-8}$ (D) $K = 10^{10}$

2. Which of the following indicator is / are suitable for titration of HC with strong base ?

(I) Phenolphthalein(8–10) (II) Thymol blue (7–9)

(III) Methyl red (4–6) (IV) Methyl orange (3–5)

(A) Only I (B) I and II (C) II and III (D) none of these

Paragraph # 2 for (Ques. Nos. 3 to 4)

Amino acid glycine (NH_2-CH_2-COOH) exist as a zwitterion in aq. solution. The K_a and K_b value of glycine are 1.6×10^{-10} ($pK_a = 9.8$) and 2.5×10^{-12} ($pK_b = 11.6$) respectively. The K_a and K_b values are for zwitterion of Amino acid with following structure $[\overset{\oplus}{N}H_3-CH_2-COO^-]$

3. What is the K_b for $-NH_2$ group in glycine

(A) 4×10^{-3}
(B) 1.6×10^{-10}
(C) 6.25×10^{-5}
(D) 2.5×10^{-12}

4. An aqueous solution of glycine has pH

(A) slightly less than 7.9
(B) slightly less than 6.1
(C) slightly greater than 6.1
(D) equal to 7.9

Paragraph # 3 for (Ques. Nos. 5 to 7)

Acid rain occurs due to combination of acidic oxides with water. During rain, $SO_2(g)$ present in atmosphere is dissolved in water to form H_2SO_3 (aq)

Given at 300 K, $K_{a_1 H_2SO_3} = 0.5$, $K_{a_2 H_2SO_3} = 10^{-7}$, R = 0.0821 Latm K^{-1} mol^{-1}

5. The solubility of $SO_2(g)$ at 1 atm and 300 K is 24.63 L in 1 litre of water. Calculate $[HSO_3^-]$ in the saturated solution of $SO_2(aq)$.

(A) 0.5 M
(B) 1 M
(C) 10^{-3} M
(D) 0.1 M

6. Degree of dissociation of water in the above solution is :

(A) 10^{-7}
(B) 1.8×10^{-9}
(C) 3.6×10^{-16}
(D) 3.6×10^{-14}

7. 100 ml 1M H_2SO_3 is titrated with 1 M NaOH solution. Calculate pH when 150 ml NaOH is added.

(A) 7
(B) 7.6
(C) 6.4
(D) 7.8